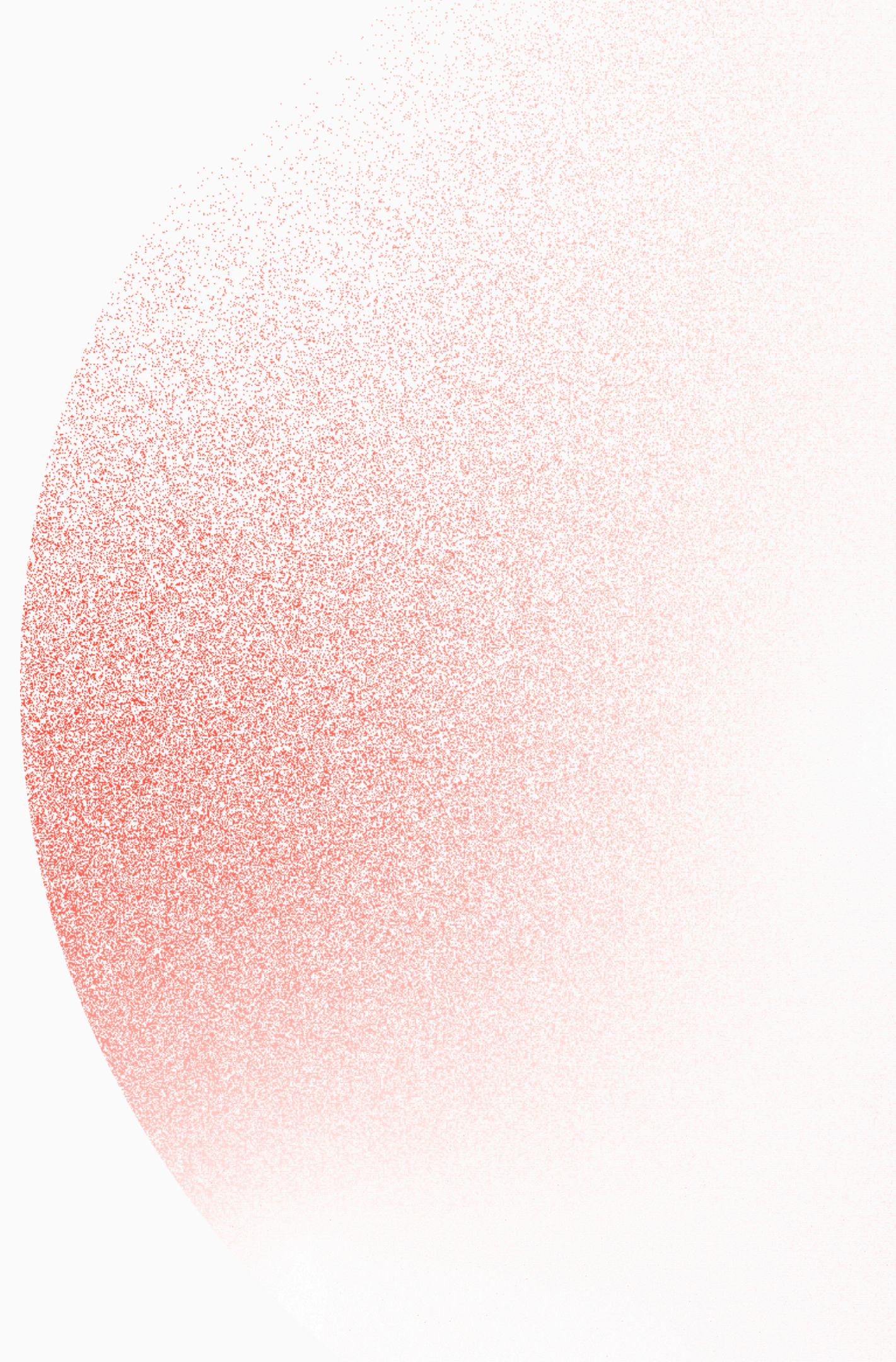


Captura de Proveniência em Pipelines de Machine Learning no Jupyter com MLflow e noWorkflow

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Pipeline

- Pipeline de aprendizado de máquina para predição da presença de doença cardíaca com base em informações clínicas dos pacientes, utilizando um modelo SVM.



MLFlow

About this run

Created at	06/23/2026, 10:58:20 PM
Created by	laet
Experiment ID	1 
Status	 Finished
Run ID	37274df40d534e13a20abab0de26a38c 
Duration	7.3s
Source	 heart_disease_pipeline.ipynb
Registered prompts	—

Datasets

 heart-disease-raw (336b7841) Raw_input

Tags

pipeline: heart_disease_pipeline **provenance_scope:** data_model_metrics
framework: scikit-learn **model_type:** SVC 

Registered models

None

```
active_run = mlflow.start_run(run_name="svm_provenance")  
  
...  
  
mlflow.end_run()
```

Runs ⓘ

MLFlow

```
raw_dataframe = pd.read_csv(dataset_path)
mlflow_dataset = mlflow.data.from_pandas(
    raw_dataframe,
    source=str(dataset_path),
    targets="HeartDisease",
    name="heart-disease-raw",
)
mlflow.log_input(mlflow_dataset, context="raw_input")
```

Data details for ● svm_provenance

1 datasets used

 heart-disease-raw (336b7841) Raw_input

 **heart-disease-raw** Raw_input

num rows: 918

num elements: 11016

Digest: 336b7841

Source type: Local

Path: /home/laet/workspaces/uff/noworkflow-ml-pipeline/datasets/heart.csv

Search fields

Name	Type
Age	long
Sex	string
ChestPainType	string
RestingBP	long
Cholesterol	long
FastingBS	long
RestingECG	string
MaxHR	long
ExerciseAngina	string

MLFlow

```
run_params = {  
    "model.kernel": svm.get_params()["kernel"],  
    "model.C": svm.get_params()["C"],  
    "model.gamma": svm.get_params()["gamma"],  
    "model.random_state": svm.get_params()["random_state"],  
    "split.test_size": 0.2,  
    "split.random_state": 0,  
    "data.feature_count": X_train.shape[1],  
}  
mlflow.log_params(run_params)
```

Parameters (7)

 Search parameters

Parameter	Value
model.kernel	rbf
model.C	1.0
model.gamma	scale
model.random_state	42
split.test_size	0.2
split.random_state	0
data.feature_count	11

MLFlow

```
run_metrics = {  
    "accuracy": float(accuracy),  
    "precision_macro": float(precision),  
    "recall_macro": float(recall),  
    "f1_macro": float(f1),  
}  
mlflow.log_metrics(run_metrics)
```

Metrics (4)

🔍 Search metrics

Metric	Value
accuracy	0.6845637583892618
precision_macro	0.684814012278801
recall_macro	0.6844144144144144
f1_macro	0.6843362632409286

MLFlow

Overview

Model metrics

System metrics

Traces

Artifacts

dataset

heart.csv

plots

confusion_matrix.png

provenance

data.json

preprocessing.txt

dataset/heart.csv

35.08KB

↓

Path: /home/laet/workspaces/uff/noworkflow-ml-pipeline/mlflow_pipeline/mlruns/1/37274df40d534e13a20abab0de26a38...

📄

Previewing the first 500 rows

⬆

Age	Sex	ChestPainType	RestingBP
40	M	ATA	140
49	F	NAP	160
37	M	ATA	130
48	F	ASY	138
54	M	NAP	150

```
{
  "source_path": "/home/laet/workspaces/uff/noworkflow-ml-pipeline/datasets/heart.csv",
  "raw_rows": 918,
  "raw_columns": 12,
  "processed_rows": 745,
  "training_rows": 596,
  "test_rows": 149,
  "target_column": "HeartDisease",
  "feature_columns": [
    "Age",
    "RestingBP",
    "Cholesterol",
    "FastingBS",
    "MaxHR",
    "Oldpeak",
    "Sex",
    "ChestPainTypeLE",
    "RestingECGLE",
    "ExerciseAnginaLE",
    "ST_SlopeLE"
  ]
}
```

MLFlow

heart-disease-svm >

Comparing 3 Runs from 1 Experiment

Visualizations

Parallel Coordinates Plot

Scatter Plot

Box Plot

Contour Plot

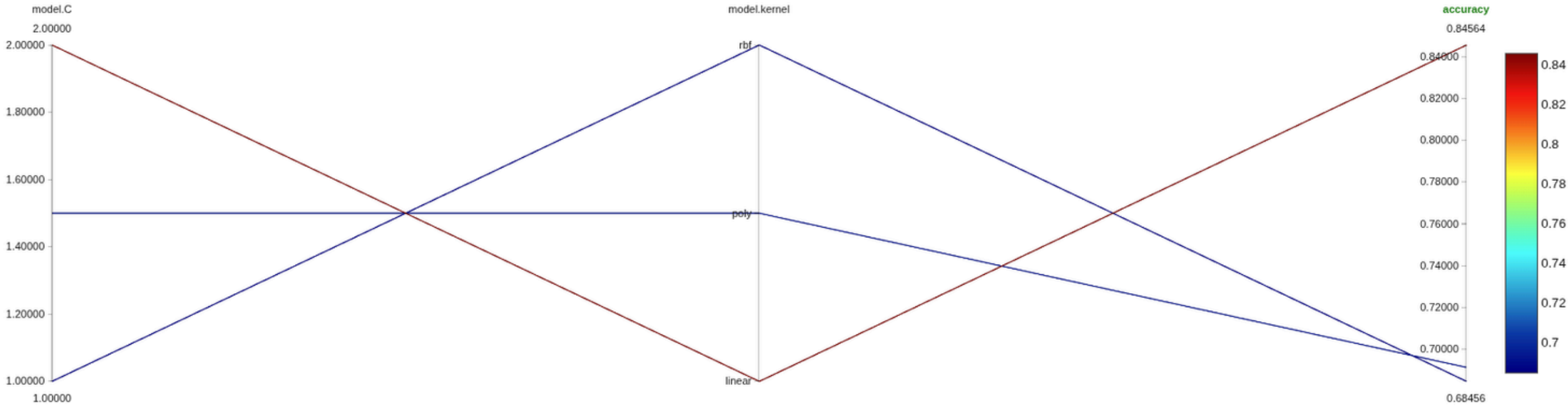
Parameters:

model.C × model.kernel ×

Metrics:

accuracy ×

Clear All



MLFlow



Run details

Run ID:	725016202db64da7ad4e34f621563ea5	d95a874200d24d79aa3510e7da055e28	68c18e0cbda1423dbb40c858730d9236
Run Name:	svm_provenance	svm_provenance	svm_provenance
Start Time:	06/20/2026, 08:49:53 PM	06/20/2026, 08:45:58 PM	06/20/2026, 08:45:41 PM
End Time:	06/20/2026, 08:50:09 PM	06/20/2026, 08:46:13 PM	06/20/2026, 08:45:50 PM
Duration:	15.1s	15.5s	9.6s

Parameters

☒ Show diff only

model.C	1.5	1.0	2.0
model.kernel	poly	rbf	linear

Metrics

☐ Show diff only

accuracy	0.691	0.685	0.846
f1_macro	0.691	0.684	0.846
precision_macro	0.691	0.685	0.846
recall_macro	0.691	0.684	0.846

MLFlow

CONCLUSÃO

- Apresenta uma proveniência em nível mais geral, centrada em **runs, parâmetros, métricas, artefatos e modelos**
- Contribui para a **reprodutibilidade** ao registrar configurações e resultados das execuções
- A captura de uma proveniência mais detalhada depende, em grande parte, da **instrumentação manual** no código

noWorkflow

Model Training

```
%load_ext noworkflow
import noworkflow.now.ipython as nip
```

SVM

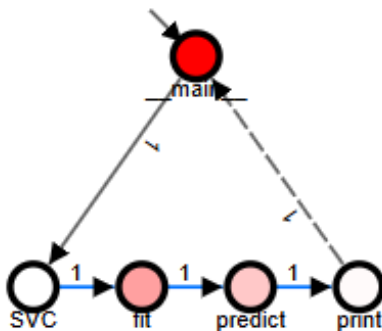
```
[25]: from sklearn.svm import SVC
```

```
[26]: %%now_run --name training --interactive
      svm = SVC(random_state=42)

      svm.fit(X_train,y_train)
      y_predicted_svm = svm.predict(X_test)
      print(y_predicted_svm)
```

```
[1 1 0 1 1 0 1 1 0 1 0 1 0 1 1 0 1 0 1 1 1 1 0 0 1 0 1 1 0 0 0 0 1
 1 1 1 1 0 0 0 1 0 1 0 1 1 0 1 0 0 1 1 0 0 1 0 0 1 1 1 1 1 0 0 1 1 0 1
 0 0 0 0 0 1 0 1 0 0 0 1 1 0 1 1 0 1 1 1 0 0 0 1 0 0 0 0 0 0 1 0 0 0 1 0 0
 1 1 1 0 0 0 1 1 1 1 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 1 1 1 0 1 0 0 1 0 1 1 0
 0]
```

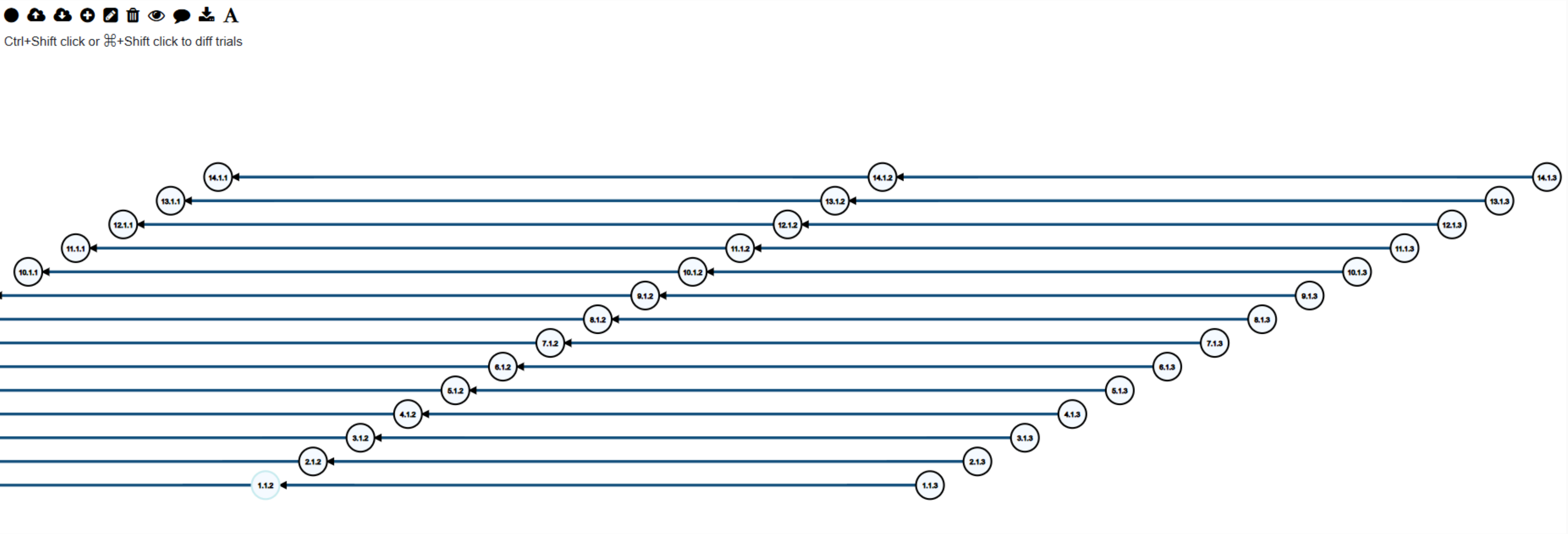
👁️ 🗨️ ⬇️ A ↔️ ⬆️



>	29	3be4d9cb-0ed3-4b91-a7ff-	new_simulation	2026-06-23 22:32:40.483996	2026-06-23 22:32:40.483996
>	30	53dc0a23-5ae6-4f87-8239-1	drop_zeroCholesterol	2026-06-23 22:32:41.411913	2026-06-23 22:32:41.411913
>	31	cc12c8cb-67f4-4736-b9c0-8	drop_sex	2026-06-23 22:32:45.938713	2026-06-23 22:32:45.938713
>	32	88c17d65-661d-4b06-972d	categorical_to_numerical1	2026-06-23 22:32:47.022907	2026-06-23 22:32:47.022907
>	33	774495e0-86e9-4940-9cdd-	categorical_to_numerical2	2026-06-23 22:32:47.719348	2026-06-23 22:32:47.719348
>	34	4a62f835-b056-46f7-be08-c	categorical_to_numerical3	2026-06-23 22:32:47.868051	2026-06-23 22:32:47.868051
>	35	5dd67982-0c54-46b5-a08c-	categorical_to_numerical4	2026-06-23 22:32:48.018431	2026-06-23 22:32:48.018431
>	36	0d509937-0cb2-40b1-9755-	drop_after	2026-06-23 22:32:48.182791	2026-06-23 22:32:48.182791
>	37	f21f2195-ab50-4725-9784-a	splitting	2026-06-23 22:32:48.964344	2026-06-23 22:32:48.964344
>	38	fe6fa89a-0e69-41ca-af66-c	training	2026-06-23 22:32:49.160791	2026-06-23 22:32:49.160791
>	39	7fcff2fc-2cc2-420a-a780-a3	prediction	2026-06-23 22:32:49.299296	2026-06-23 22:32:49.299296
>	40	6a8cd694-0346-43cd-aaf8-4	example1	2026-06-23 22:32:49.406361	2026-06-23 22:32:49.406361
>	41	59ac8434-2de5-4391-85f1-1	example2	2026-06-23 22:32:49.546671	2026-06-23 22:32:49.546671
>	42	8c2946b0-e439-40ed-908a-	metrics	2026-06-23 22:32:49.716976	2026-06-23 22:32:49.716976

noWorkflow

```
now vis -b
```



noWorkflow

CONCLUSÃO

- Útil para acompanhar e comparar execuções de células do notebook, capturando a ordem e valores das instruções dentro da célula de maneira automática.
- Necessita de pouca configuração para capturar os dados, sendo de fácil uso.
- Falta de controle sobre o escopo da coleta gera um volume alto de dados redundantes, coletando informações de funções de visualização (prints/plots) e gerando um overhead em células de processamento intenso.

MLFlow x noWorkflow

COMPARAÇÃO

Aspecto	MLFlow	noWorkflow
Granularidade da captura	Mais geral , centrada no experimento	Mais detalhada , centrada na execução das células
Captura	Parâmetros, métricas, artefatos e modelos	Ordem, instruções e valores durante a execução
Instrumentação	Necessida de logs manuais	Captura automática
Tipo de rastreabilidade	Ajuda a identificar qual configuração gerou determinado resultado	Ajuda a entender como o resultado foi produzido durante a execução

MLflow



noWorkflow